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Local Recurrence After Uveal Melanoma Proton Beam Therapy: Recurrence Types and Prognostic Consequences

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Summary

We have studied 61 cases of uveal melanoma local recurrences after proton beam therapy. Our local recurrence rate was 6.1%. These local control failures were divided into 4 recurrence types. The large tumoral diameter has been the sole statistical significant recurrence risk factor found. The overall survival at 10 years was 68.7% for recurrence-free patients and 43.1% for local recurrence patients. The multivariate analysis has shown a better prognosis for marginal recurrences.

Purpose: To study the prognosis of the different types of uveal melanoma recurrences treated by proton beam therapy (PBT).

Methods and Materials: This retrospective study analyzed 61 cases of uveal melanoma local recurrences on a total of 1102 patients treated by PBT between June 1991 and December 2010. Survival rates have been determined by using Kaplan-Meier curves. Prognostic factors have been evaluated by using log-rank test or Cox model.

Results: Our local recurrence rate was 6.1% at 5 years. These recurrences were divided into 25 patients with marginal recurrences, 18 global recurrences, 12 distant recurrences, and 6 extrascleral extensions. Five factors have been identified as statistically significant risk factors of local recurrence in the univariate analysis: large tumoral diameter, small tumoral volume, low ratio of tumoral volume over eyeball volume, iris root involvement, and safety margin inferior to 1 mm. In the local recurrence-free population, the overall survival rate was 68.7% at 10 years and the specific survival rate was 83.6% at 10 years. In the local recurrence population, the overall survival rate was 43.1% at 10 years and the specific survival rate was 55% at 10 years. The multivariate analysis of death risk factors has shown a better prognosis for marginal recurrences.

Conclusion: Survival rate of marginal recurrences is superior to that of the other recurrences. The type of recurrence is a clinical prognostic value to take into account. The influence of local recurrence retreatment by proton beam therapy should be evaluated by novel studies.

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Introduction

Uveal melanomas (UMs) are the main adult intraocular tumor. The local control by proton beam therapy (PBT) of these tumors is good in the different series results (1-9). Many studies showed higher death and metastatic rates in case of local control failure (6-8, 10, 13). However, different types of recurrence are possible, with various definitions and descriptions in the literature (2, 3, 7, 8, 13, 14). Moreover, the recurrence type is seldom specified in most studies.

Methods and Materials

Patients

All our patients have been treated by PBT according to the technique previously described after positioning 4 tantalum rings on the sclera after UM localization by transillumination (3, 15). The safety margin was usually 2.5 mm. A lower safety margin was used for some posterior pole tumors to increase functional results (3). This is made possible by our beam ballistic properties with a short distal penumbra. The total dose delivered was 60 cobalt Gray equivalents in 4 fractions. After undergoing treatment, patients were regularly followed by their local ophthalmologist. A thorough examination was performed by our physicians at each half-yearly check-up as for the initial work-up with fundus examination after dilatation, retinography, B-mode ultrasound, or ultrasound biomicroscopy, fluorescein angiofluorography, computed tomography scan, and/or magnetic resonance imaging. Patients were examined in our clinic every 6 months for 2 years and then every year for 5 years and at 7, 10, and 15 years. They were checked earlier in case of problem or doubt on treatment efficiency. In case of recurrence, we firstly look at initial the images to eliminate errors in the initial treatment planning. Liver ultrasonography or tomodensitometry was performed to detect metastases every 6 months for 5 years and then every year for 10 years.

We studied the local recurrence cases in a total of 1102 UMs treated by PBT between June 1991 and December 2010. For statistical analysis, the patients have been divided into 2 subgroups (ie, patients controlled by their PBT and patients with a local control failure).

Description of the different types of local control failure

For this study, we have divided the local control failures according to 4 possible types (3, 11). The "marginal recurrences" are a tumor growth on the lesion margin with initially a lateral flat extension of the UM. The "global recurrences" are a persistent growth or a regrowth of the tumor in all their dimensions. The "extrascleral extensions" are the development of the tumor in the orbit through the sclera. The "distant recurrences" are a new tumoral localization inside the eyeball with a free choroidal interval of at least 5 mm outside the irradiated area. The irradiated area that receives more than 90% of the dose is defined as the "clinical target volume" extended by a standard 2.5-mm margin also referred to as "planning target volume."

A tumoral growth was only considered as a recurrence if it kept progressing after the first 6 months and/or after 2 consecutive examinations. Most recurrences are tumoral regrowths after an initial stabilization or tumor shrinkage except for the distant and the extrascleral recurrences. In case of a doubt, the patients were seen earlier (ie, 3 months later) in order to eliminate false recurrences (7), namely subretinal and/or tumoral hemorrhage, lipidic exudation, or edema following proton beam irradiation (16).

Statistical methods

All categorical data were described using figures and percentages.

Quantitative data were presented using medians and ranges or means and standard deviations. Censored data were described using Kaplan-Meier estimation, including the number of patients, number of events, median survival, and a 95% confidence interval (CI). Whenever the information was not available, the status was coded as missing data. Statistical analyses were 2-sided and performed using R-2.5.0 software for Windows. All censored data intervals were calculated from the date of the first intraocular tumor treatment to the event (patient death, lost to follow-up, or metastatic or local recurrence).

Univariate analyses

Statistical comparisons were performed using the chi-square test or Fisher's exact test for categorical data, Student t test or Wilcoxon test for quantitative data, and log-rank test for censored data.

Multivariate analyses

Multivariate analyses were performed by creating a Cox proportional hazard model or a logistic regression. Choice of the final model was made by performing a backward stepwise model selection by exact Akaike information criterion. Usually, all variables associated with a P value $<.10$ on univariate analysis were included in the model. Colinearity between variables was evaluated using the Pearson r correlation coefficient value between all variables entered in the model. Whenever r was >0.30, 1 of the 2 variables was considered redundant and was removed from the model.

Results

We have found 61 cases of local recurrences on a total of 1102 UMs treated by PBT. Seven patients were lost to follow-up; they belonged to the recurrence-free patients (RFPs) group. Our RFP rate was 93.9% ($\pm$0.9%) at 5 years, 91.3% ($\pm$1.2%) at 10 years, and 90.6% ($\pm$1.4%) at 15 years. A total of 77% of our local recurrences occurred within the first 3 years of the treatment. The last recurrence was observed 16 years after the PBT.

Qualitative and quantitative tumoral features concerning our series are displayed Tables 1 and 2. There was no statistical difference between local recurrence patients (LRPs) and recurrence free patients for the gender, the affected eye, the localization, and the 7th TNM classification. There was no statistical difference between LRPs and RFPs regarding the age and the initial tumoral thickness. Conversely, there was a statistical difference between LRPs and RFPs concerning the initial tumoral

Table 1 Univariate analysis of qualitative recurrence risk factors

Qualitative variables	Modality	Nb patients with recurrences/Nb patients (recurrences%) [CI 95%: minimum-maximum]	*P* value
Gender	Male	29/498 (5.8%) [1.5-10.1]	.80993
	Female	32/603 (5.3%) [1.3-9.3]	
Affected eye	Right	29/556 (5.2%) [1-9.3]	.72546
	Left	32/544 (5.8%) [1.7-10]	
Localization	Pre-equatorial	14/188 (7.5%) [0.5-14.6]	.57041
	Equatorial	20/427 (4.7%) [0-9.4]	
	Retro-equatorial	27/485 (5.6%) [1.2-10]	
Initial nevus	No	60/1084 (5.5%) [2.6-8.4]	.63684
	Yes	1/17 (5.9%) [0-29]	
Optic nerve involvement	No	46/912 (5%) [1.8-8.2]	.14855
	Yes	15/187 (8%) [1-15]	
Initial retinal detachment	No	36/679 (5.3%) [1.6-9]	.49232
	Peritumoral	8/161 (5%) [0-12.8]	
	Inferior	15/245 (6.1%) [0-12.2]	
	Subtotal	2/12 (16.7%) [0-43]	
	Total	0/4 (0%) [0-0]	
Intraocular hemorrhage	No	53/960 (5.5%) [2.4-8.6]	.79198
	Yes	6/88 (6.8%) [0-17]	
Initial extrascleral extension	No	60/1079 (5.6%) [2.6-8.6]	.95895
	Yes	1/21 (4.8%) [0-26]	
Vortex vein involvement	No	57/1027 (5.6%) [2.6-8.6]	.83328
	Yes	4/74 (5.4%) [0-16.8]	
Ciliary body involvement	No	41/801 (5.1%) [1.7-8.5]	.39434
	Yes	20/300 (6.7%) [1.1-12.3]	
Iris root involvement	No	52/1039 (5%) [2-8]	.0038
	Yes	9/62 (14.5%) [2.8-26.2]	
7th TNM	1	11/296 (3.7%) [0-9.4]	.33097
	2	26/464 (5.6%) [1.1-10.1]	
	3	20/292 (6.8%) [1.2-12.4]	
	4	4/50 (8%) [0-21.6]	

The mention of a few tumoral features are missing in our database, notably at the very beginning of the proton beam therapy data, which explains that if we add the variables, we do not reach 1102 in all cases.

diameter and the follow-up. The mean initial diameter was higher and the follow-up longer for LRPs.

Our recurrences were divided into 25 marginal recurrences (41%), 18 global recurrences (29.5%), 12 distant recurrences (19.7%), and 6 extrascleral recurrences (9.8%). The initial UM localization of global and marginal recurrences were mainly retro-equatorial. As for the extrascleral and distant recurrences, they were mainly pre-equatorial.

Table 2 Univariate analysis of quantitative recurrence risk factors

Quantitative variables	Patient without recurrences Median [min-max]/patient number Or for safety margin variable: case number/patient number (%) [CI 95% min-max]	Patient with recurrences Median [min-max]/patient number Or for safety margin variable: case number/patient number (%) [CI 95% min-max]	*P* value
Age (years)	65.29 [14.74-92.72]/1034	60.89 [27.09-85.29]/61	.13867
Follow-up (years)	3.93 [0-17.84]/1034	5.11 [0.22-17.86]/61	.03092
Initial diameter (mm)	11.11 [3.6-24.14]/1030	13.4 [7-20]/61	.00092
Initial thickness (mm)	5 [1-15]/1037	5 [2.5-12]/61	.72615
Safety margin $<$ 1mm	30/35(85.7%) [79.3-92]	5/35(14.3%) [0-30]	.0457
Safety margin $<$ 2mm	392/418(93.8%) [92.6-95]	26/418(6.2%) [1.5-11]	.38445
Tumoral volume (mL)	0.55 [0.01-1.9]/920	0.5 [0.07-2.3]/52	.00276
Distance from macula (mm)	3.1 [0-25.2]/1018	3.1 [0-20.5]/61	.50385
Distance from papilla (mm)	4 [0-23.5]/1017	3.7 [0-21.4]/59	.83011

The mentions of a few tumoral features are missing in our database notably at the very beginning of the PBT data. This fact explains that if we perform the addition of the variables, we do not reach 1102 in all the cases.

Identification of recurrence risk factors

In our univariate analysis, 4 statistical significant recurrence risk factors have been identified: initial tumoral diameter (larger in LRPs), initial tumoral volume (smaller in LRPs), the iris root involvement, and safety margin inferior to 1 mm. Conversely, the use of a 2-mm safety margin was not a risk factor. Moreover, there were no significant statistical differences for the following features: initial tumoral thickness, distances from papilla and from macula, initial exudative retinal detachment regardless its size, tumoral localization, and ciliary body involvement (except tumors with iris root involvement as previously mentioned). The data are shown in Tables 1 and 2.

In the multivariate analysis of the recurrence risk factors, the large tumoral diameter has been the sole statistical significant recurrence risk factor found with a P value of .0012. The involvement of ciliary body ($P=.316$), the use of a safety margin inferior to 1 mm ($P=.081$) and tumoral volume ($P=.175$) were not found as recurrence risk factors.

Patient consequences of recurrences

Regarding overall survival

For the RFP population, the overall survival (OS) was 81.2% at 5 years (CI 95%: 78.3-84.1) and 68.7% at 10 years (CI 95%: 64.5-73.2). For the LRP population, the OS was 74.1% at 5 years (CI 95%: 63.2-86.8) and 43.1% at 10 years (CI 95%: 29.3-63.3). In our series, we have noted a lower OS for LRPs compared with RFPs with a significant statistical difference ($P=.00123$).

Regarding disease-specific survival

For the RFP population, the disease-specific survival (DSS) was 89% at 5 years (CI 95%: 86.6-91.3) and 83.6% at 10 years (CI 95%: 80.3-87.1). For the LRP population, the DSS was 80.3% at 5 years (CI 95%: 69.9-92.1) and 55% at 10 years (CI 95%: 39.6-76.2). The DSS is lower for LRPs compared with RFPs with a significant statistical difference as shown on Figure 1.

For survival according to the recurrence types

Survival Kaplan-Meier curves showed a better prognosis for RFPS compared with the other recurrence patients (Fig. 2) and did not show any prognostic statistical significance between RFPs and marginal recurrence patients ($P=.208$). The DSS at 5 and 10 years were 89% (CI 95%: 86.6-91.4) and 83.6% (CI 95%: 80.3-87.1) for RFPs, 86.4% (CI 95%: 73.1-100) and 72.0% (CI 95%: 53.1-97.4) for marginal recurrence patients, and 75.7% (CI 95%: 61.3-93.6) and 34.2% (CI 95%: 13.7-85.2) for the other recurrence patients respectively. If we focus on the other recurrence subtypes, the DSS at 5 years was 74.1% for distant recurrence patients (CI 95%: 52.6-100), 84.8% for global recurrence patients (CI 95%: 67.3-100), and 40% for extrascleral recurrence patients (CI 95%: 9.3-100). The median survival for each recurrence type was 15.3 years for marginal, 6.2 years for global, 6.5 years for distant, and 3.9 years for extrascleral recurrences.

The multivariate analysis of metastatic risk death related to the tumor has found 2 independent death risk factors: a large tumoral diameter and, importantly, the recurrence type (ie, the local recurrences except marginal recurrences). Thus, the global, the distant and the extrascleral recurrences have showed a high risk of metastatic death unlike marginal recurrences (Table 3).

In our series, 24 patients with PBT local control failures benefited from a second PBT. These patients were divided into 1 extrascleral, 16 marginal, 4 global, and 3 distant recurrences. A total of 29 patients presenting with recurrences underwent an enucleation. They were divided into 5 marginal, 14 global, 7 distant, and 3 extrascleral recurrences. Two patients were treated with argon laser for their recurrences. The other recurrences were not treated because of their metastatic status. The OS in reirradiated LRPs was better as compared with enucleated LRPs with a slight statistical significance (Fig. 3). The OS of reirradiated patients was 87.5% at 5 years (CI 95%: 75.2-100) and 61.4% at 10 years (CI 95%: 41.8-89.9). The OS of enucleated patients was 73.1% at 5 years (CI 95%: 56.4-94.8) and 22.3% at 10 years (CI 95%: 0.07-67.1). Concerning the marginal recurrences retreated by PBT, 14/16 patients

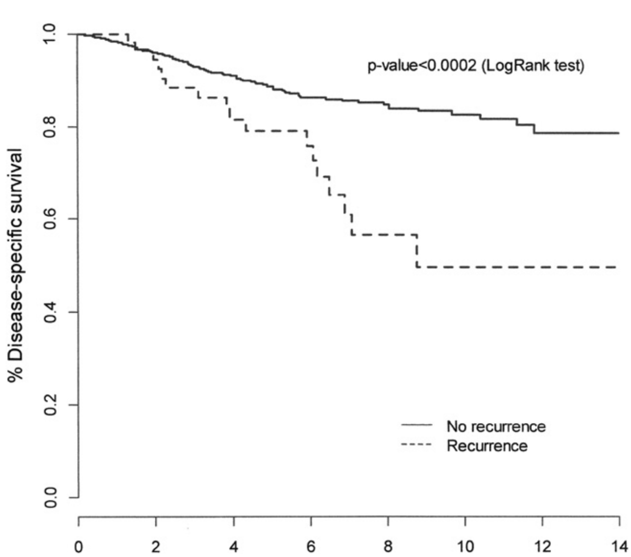

Fig. 1. Kaplan-Meier curves for specific survival according to local recurrence.

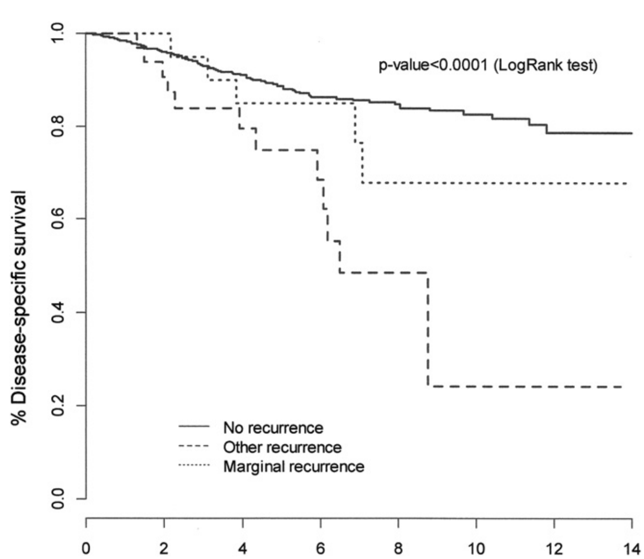

Fig. 2. Kaplan-Meier curves for specific survival comparing patients without recurrence versus marginal recurrence patients and versus patients with other local recurrence types.

Table 3 Multivariate analysis of death risk factors related to the tumor

	Univariate analysis			Multivariate analysis		
	Hazard ratio	CI 95%	*P* value	Hazard ratio	CI 95%	*P* value
Age (y)	1.01	[0.99-1.02]	.20	Not included*		
Tumor thickness (mm)	1.15	[1.08-1.21]	$<10^{-5}$	Not included*		
Tumor diameter (mm)	1.20	[1.15-1.26]	$<10^{-5}$	1.20	[1.13-1.26]	$<10^{-5}$
Tumor volume (mL)	1.09	[1.01-1.17]	.030	1.02	[0.90-1.14]	.79
Recurrence						
No recurrence	1	-	-	1	-	-
Marginal recurrence	1.7	[0.74-3.9]	.216	1.63	[0.70-3.77]	.25
Other recurrences	3.4	[1.9-6.2]	$<10^{-4}$	3.23	[1.75-5.97]	.0002
Recurrence treatment						
Second proton therapy	1	-	-	Not included*		
Enucleation	2.6	[0.86-8.2]	.09	Not included*		
Ciliary body involvement						
No	1	1	-	1	-	-
Yes	1.70	[1.20-2.50]	.004	1.02	[0.66-1.57]	.93

Colinearity between variables was evaluated using the Pearson *r* correlation coefficient between all variables entered in the model. If $r>0.30$, one of the two variables is considered as redundant and must be removed from the model. This is the case for tumor thickness and tumor diameter ($r=0.57$). Only tumor diameter was introduced into the model. If death was not due to cancer or if the patient was lost to follow-up, data were censored at the date of his or her last known contact.

* Variable not included in the model because it is not significant in univariate analysis or it is correlated with other covariate.

(87.5%) achieved a tumoral local control. At the time of retreatment, 11/16 had a useful vision better than 20/200, and 5/11(45%) maintained it at their last controls.

Discussion

Our local control rate of 93.9% and 91.3% at 5 and 10 years, respectively, was close to that of Egger (7) (ie, 95.8% and 94.8%). Our local recurrence rate is similar to those of the other series, which range from 2 to 8.4% at 5 years (1-9). Our last recurrence was observed at the 16th year of follow-up (192 months). In the literature, the very last recurrences were observed at 11.3 years for Gragoudas (2) and 130 months for Egger (7).

Recurrence types

The margin recurrences were mentioned by many authors (2, 3, 7, 11, 14). They might be due to an "insufficient radiation dose at the tumoral border following errors in treatment planning or in the irradiation delivery" as mentioned by Desjardins (16). They were often noted for posterior localized tumors or for tumors involving ciliary body. Our global recurrences were the "uncontrolled tumors" of Gragoudas (2), the "vertical recurrences" of Harbour (14), the "continued growth within the original target volume" of Egger (7), or the "tumor regrowth within the irradiated area" of Marucci (11). They might be due to a tumor radioresistance. Concerning "distant recurrences," they were rare and not clearly described. We have defined them as a new tumoral localization with a free choroidal interval that is the double of the usual 2.5-mm safety margin. For Marucci and Egger (7, 11), they are recurrences "completely outside the irradiated area" without any precision. These distant recurrences are mainly observed in ciliary tumors notably if there is an iris root involvement. This fact causes a slight inconsistency, with our other results showing that the localization and ciliary tumors are not recurrence risk factors. These distant recurrences might be due to the melanoma cell spreading throughout the anterior chamber or to the extension of the melanoma along the ciliary body (like for "ring melanoma" noted by Gragoudas). However, they were also noted for melanomas strictly localized at the equator or the posterior pole. They may be due to the migration of tumor cells into exudative retinal detachment as mentioned by Desjardins (16) who recommends a careful examination of the inferior periphery at each post irradiation control.

The main recurrence type of our series was the marginal recurrence with 41%, as in the scarce studies specifying the recurrence types (eg, 20%-70% of recurrences; 2, 3, 7, 8, 13, 14).

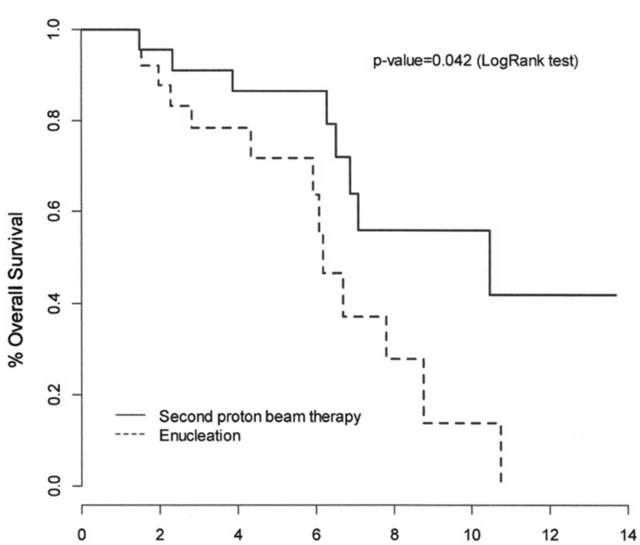

Fig. 3. Kaplan-Meier curves comparing overall survival of patients treated by enucleation versus patients reirradiated by proton beam therapy.

Identification of recurrence risk factors

In our statistical analyses, like many authors (2-7, 9, 14), we have found the large initial tumoral diameter as a statistical significant recurrence risk factor. We can assume that large and thin melanomas are more likely to recur because of the difficulty in defining the lateral tumoral limits properly. Moreover, extra large and flat type melanomas referred to as "diffuse," would be of more aggressive histological type (17).

In our univariate analysis, we have found a low statistical significant risk factor for safety margins inferior to 1 mm but not when they were inferior to 2 mm. Egger (7) has previously noted that the safety margin reduction was a risk factor. It means that the lower the safety margins are the higher the local recurrences will be. Yet, this risk factor has not been revealed in our recurrence factor multivariate analysis.

The distances from macula and papilla were not local recurrence risk factors in our series unlike Quivey (12), Char (18), Karlsson (19), Harbour (14), and Kodjikian (8). However, most of these authors are using brachytherapy except Kodjikian. Conversely, Dendale (6) has noted that distance from macular area could be a prognostic factor of local control.

Unlike Desjardins (16) and our results, Gragoudas (2) and Egger (7) have found the initial ciliary body tumoral involvement as a significant risk factor of local control failure. It might be due to the fact that we are using an increase in safety margins whenever UMs involve the ciliary body, the transillumination of tumor limits being difficult for this area.

Patient consequences of recurrences

As for the survival, the literature data, like our study, showed that a local control failure was unfavorable in terms of metastasis and death (2, 3, 6-10, 14). Gragoudas (2) found that the relative risk of metastasis-related death was 4.1 for LRPs as compared with RFPs. Egger (7) found a decrease in the survival rate from 72.6% for RFPs to 47.5% for LRPS. To compare with our study, we have noted a drop in the OS at 10 years from 68.7% for RFPs to 43.1% for LRPs.

In our multivariate analysis, the independent death risk factors are large tumoral diameter and the recurrence types. Indeed, the death risk is higher with global, distant, and extrascleral recurrences and not with marginal recurrences. The large tumoral diameter factor is often found in all the studies on UM PBT (2, 3, 5-9). To our knowledge, only Harbour (14) has previously studied the death rate according to the type of recurrence. Similarly to our study, he has showed that mortality was lower with marginal recurrences (19%) than with global recurrences (49%). Yet, 51 patients of his series underwent brachytherapy and only 15 helium ion therapy.

In addition, we have not noted any statistically significant difference between specific survival of marginal recurrence patients and RFPs. The recurrence type might be an important prognostic factor to take into account in the local control failure treatment.

For the survival rate, which was lower with LRP population than with RFP population, the explanations were: first, the unfavorable tumoral diameter that is a risk factor of both metastases and local recurrences; and second, a more aggressive histological type. Moreover, to understand the survival difference rates between marginal recurrences and the others, it seems impossible to determine whether metastases are directly correlated with the uncontrolled tumoral cells of the recurrence or the aggressiveness of the initial tumor (histologic and/or genomic).

In our study, we consider 3 hypotheses to account for the best results in terms of survival in the patients with marginal recurrences. The first is an imperfect initial treatment planning corrected by a second course of PBT without any loss of chance for the patient. The second is a protective effect over the metastatic development and mortality by the second PBT course. Indeed, Marucci (11) has shown that mortality and distant metastases are lower in recurrences treated by a second course of PBT than in recurrences treated by enucleation, as mentioned in our study. However, there may have been a statistical bias as the patients with poor prognostic factors were more likely to undergo enucleation than reirradiation. Immune mechanism such as the "abscopal effect" described by Demaria (20) or angiogenic mechanisms hypothesized by Gragoudas (2) might also be possible. The third explanation could lie in the fact that marginal recurrences would be a less aggressive subtype or phenotype than the other recurrences. Unfortunately, for this study, we still do not have any genomic tumoral analyses.

Strategy suggestions

To prevent local recurrences, large UMs up to 20 mm in initial diameter should not have a conservative treatment. We should not use safety margin inferior to 1.5 mm even in specific cases. In ciliary body tumors with iris root involvement, we should be careful with the cell spreading through the anterior chamber yielding distant recurrences in the iridocorneal angle. The physician must always perform a control of the tumoral dimensions right before the PBT onset in view of detecting possible modifications since the last examination. At the same time, the physician will be in a position to review the treatment planning to detect reconstruction errors causing local control failure.

For the management of local control failure, marginal recurrences might be reirradiated as often as possible. Indeed, these recurrences have a good disease-specific survival and their retreatments seem to get satisfactory local control rates and functional results.

Conclusions

This retrospective analysis indicates that the survival rates of marginal recurrences are better than those of the other recurrence types. Moreover, these rates are close to those of the RFPs. Consequently, the type of recurrence is an important clinical prognostic value to take into account. However, when we look at the conclusion of Marucci study (14), it might be a statistical bias because the marginal recurrences are mainly retreated by PBT and the others by enucleation.

Collaboration studies among different centers interested in this topic seem important in order to obtain bigger samples giving more significant statistical analyses. Thus, we will be able to assess the prognosis of each recurrence type, the beneficial effect of reirradiation by PBT, and the genomic characteristic of the different recurrences.

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